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Optima Academy Online

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Teacher Manual

Probability & Statistics Honors · Florida 1210300 · Grades 9–12 · 2026–2027

Facilitator copy. Contains answer guidance, rubrics, grading thresholds, and internal standards alignment.
Do not publish to students or attach to any student-facing Canvas item.

SECTION 0

0. How to use this manual

This manual documents the course as built: Semester 1 (Modules 1–6, Weeks 1–17, plus the Week 18 finals week) and Semester 2 (Modules 7–12, Weeks 19–35, plus the Week 36 finals week) are both fully constructed. Semester 2 pages were written in October 2026 and carry open items the build could not settle (the Excel click-paths, the Desmos entries and the VR staging have not been tested in live tools or in a headset); anything the build does not yet settle is marked **[VERIFY]**, and the Semester 2 widgets must be pushed to the course's GitHub Pages repo before Semester 2 opens.

Read Sections 1–3 before your first week — they tell you what students actually see. Set up your gradebook from Section 5. Keep Sections 6, 7, and 9 open during the term: they are the rubrics, the per-item grading notes, and the weekly announcement scripts. Section 10 tells you where your heavy weeks are; Section 14 is the standards map; Section 15 is the guide for running the course as a live class meeting four days per week.

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SECTION 1

1. How this course works

The course is fully asynchronous and self-contained: probability and decision theory in Semester 1, statistics in Semester 2. There is no textbook for students to buy or open — all reading is embedded on the lesson pages, adapted into Optima voice from two open-source texts that are credited once, in the Module 1 intro wrapper, and never named again on student pages. A student needs only Canvas, the embedded simulators, the VR campus, the free Desmos Scientific Calculator, and (in Semester 2) Excel.

A typical lesson (walk-through of Lesson 1.1, “The Monty Hall Problem”)

Every lesson page follows the same anatomy, so students always know where they are:

- **Header & metadata strip** — course banner, module/quarter, lesson number, module virtue theme. The Reading cell says “Embedded in this lesson” — a deliberate signal that the page is complete in itself.
- **Today’s Focus** — the hook (“a game show puzzle that fooled professors of mathematics”) and a time estimate (about 60 minutes).
- **Objectives** — 2–4 “I will be able to...” statements aligned to the module’s benchmarks (codes tracked internally only; see Section 14).
- **Materials** — always: notebook and pencil, Desmos in another tab, and any embedded simulator.
- **Content sections with commit-then-reveal** — the exposition asks the student to write an answer *before* a collapsible “show me the verdict” panel opens. Worked examples are computed step by step, with multiple representations.
- **Embedded widget** (where the topic supports it) — an interactive simulator (Monty Hall, coin streaks, Bayes updater, Galton board...) served from the course’s GitHub Pages widget repo and embedded as an iframe. Ungraded, but activities reference the runs students make in it.
- **Recurring callouts** — a gold *In the Real World* application; a *Calculator Skills* box with exact Desmos (S2: also Excel) button paths; an *AI Training* box (see Section 3); and, on many lessons, a blue *history of mathematics* mini-lesson (Pascal–Fermat, Bayes, Bernoulli, Galton, Fisher...).

- **Activities** — 2–4 sequenced tasks (narrate your reasoning, worked-then-independent practice, error analysis, an applied problem) that feed directly into the paired assignment.

A typical week

Segment	What the student does	Graded?	Tier
Lesson 1 (≈60 min)	Reads/works the page, runs the widget, completes activities in notebook	No (feeds assignment)	4
Assignment 1 (10 pts)	Submits notebook photo/scan or typed work; practice set scored on complete, honest work	Yes — 2-row rubric (6 completeness / 4 work shown)	1
Quiz 1 (5 pts, 5 questions)	Auto-graded; multiple choice / numeric; every item also feeds the semester-final bank	Yes — auto	1
...then the same lesson → assignment → quiz cycle twice more (Lessons 2 and 3). Weekly totals: 3 lessons, 3 assignments (30 pts), 3 quizzes (15 pts).			

On top of the weekly cycle, each **module** (2–3 weeks) carries: an **intro wrapper** (ungraded survey quiz, first item), an early **VR activity** (5 pts, completion), a culminating **VR lab** (15 pts, rubric — the module’s deep-graded item), and an **outro wrapper** (graded survey, 2 pts completion, last item).

Major assessments

Assessment	Module · Week	Pts	Tier	Teacher grading cost (per cohort pass)
VR Lab — Monty Hall at Scale	M1 · Wk 3	15	3	The six VR labs are the only deeply graded items in Semester 1. Rubric is 3 clicks + a one-line comment (≈2–3 min/student). Spread each lab’s grading across the following week.
VR Lab — The Counting Gallery	M2 · Wk 6	15	3	
VR Lab — The Screening Ward	M3 · Wk 9	15	3	
VR Lab — The Galton Drop	M4 · Wk 12	15	3	
VR Lab — The Risk Room	M5 · Wk 15	15	3	
VR Lab — The Betting Parlor	M6 · Wk 17	15	3	
Semester 1 Review Activity	Finals · Wk 18	10	1	Completion-style scan
Semester 1 Final Exam (50 questions)	Finals · Wk 18	50	1	Auto-graded (bank draw; see Section 8)

Semester 2 mirrors this shape. Its six VR labs, its review activity and its final are:

Assessment	Module · Week	Pts	Tier	Teacher grading cost (per cohort pass)
VR Lab — The Chart Room	M7 · Wk 21	15	3	Rubric is 3 clicks + a one-line comment ($\approx$ 2–3 min/student). Spread each lab's grading across the following week.
VR Lab — Walking the Distribution	M8 · Wk 24	15	3	
VR Lab — Inside the Bell	M9 · Wk 26	15	3	
VR Lab — The Scatter Field	M10 · Wk 29	15	3	
VR Lab — The Sampling Orchard	M11 · Wk 32	15	3	
VR Lab — The Gallery of Samples	M12 · Wk 35	15	3	
Semester 2 Review Activity	Finals · Wk 36	10	1	Completion-style scan
Semester 2 Final Exam (50 questions)	Finals · Wk 36	50	1	Auto-graded (bank draw; see Section 8)

Semester 2 also has six early VR activities (5 pts each, completion: Data Safari Wk 20, The Outlier Hunt Wk 22, z-Score Lineup Wk 25, Guess the Correlation Gallery Wk 28, The Bias Museum Wk 31, The Interval Walk Wk 33) and eleven **Excel workbook uploads** (see Section 7). **Two Semester 2 assignments carry teacher-graded written work beyond the VR labs:** Assignment 32.3 (design-a-study workshop; 7 of its 10 pts are a rubric-scored design) and Assignment 35.3 (capstone critique; 7 of its 10 pts are a rubric-scored written critique with an AI-comparison reflection). Plan these into Weeks 32–33 and 35–36; Assignment 32.1 also has two items that need a teacher check.

SECTION 2

2. Ungraded interactives & the module wrappers

The interactives cost you nothing

Simulators (Monty Hall, dice roller, Bayes updater, expected-value games, and about forty more across the year) and the commit-then-reveal panels are **formation by practice, not gradable work**. They produce no submission, need no grading, and generate no missing-work flags. A student who skips them shows up indirectly — in quiz performance and thin assignment work — not as a gradebook hole. Do not chase widget completion; chase the quiz and assignment signal it produces.

Wrappers are survey quizzes — and triaging them is your highest-leverage 20 minutes

Each module opens and closes with a Canvas Classic Quiz in **survey mode** (intro: ungraded; outro: graded survey, 2 pts completion). They are quizzes, not pages, so responses land as **sortable cohort data**. The intro captures the module's essential question (first answer), an objectives acknowledgment, a group-work opt-in, and a confidence baseline. The

outro captures the refined essential-question answer, a **confidence rating (the early-warning signal)**, a “which topic deserves another pass” vote, a free-response “what remains unclear,” and the module’s virtue reflection.

The move: within a few days of each module’s close, spend 15–20 minutes in the outro analytics. Sort the confidence question and flag every “Shaky” and “Lost” for outreach; tally the “another pass” vote to pick your reteach topic; skim “still unclear” in batches for the specific step that lost people. Feed what you find into the next weekly announcement, one targeted reteach note, and 3–5 individual check-in messages. Nothing else you do in this course buys more per minute.

The Module 1 intro wrapper’s description also carries the course’s single source-text disclosure. Keep that paragraph intact — it is the one place the open-source anchor texts are credited, by design. Do not copy the internal “teacher triage notes” from the wrapper build files into Canvas.

SECTION 3

3. AI & VR in this course (what students experience)

AI: every lesson trains supervised skepticism

Every lesson carries one **AI Training callout** on a fixed pattern: *prompt* → *predict* → *verify*. The student crafts a prompt for an AI tool, predicts or computes the answer independently, then grades the AI’s response against their own work. The course deliberately parks these tasks where AI tools stumble — conditional probability, base rates, “is this coin due,” what a p-value is *not*, correlation-vs-causation — so students catch real errors, not staged ones. The posture is awareness and verification: no specific AI tool is required, and the callouts never outsource the reasoning. Do not add new AI requirements on top of these; when students ask, point them to the syllabus rule (AI may support learning; it may never do the assigned thinking, and its use is disclosed).

VR: two solo assignments per module

Each module has an early **VR activity** (5 pts, completion — e.g., arranging the 36-outcome dice grid, tallying urn draws) and a culminating **VR lab** (15 pts, rubric — e.g., enacting all three Monty Hall worlds at room scale, walking a 1,000-figure screening ward). Sessions are 20–30 minutes, fully solo, and every one ends in a single Canvas submission: a snapshot or short spatial recording plus a brief reflection. Each VR lab page ends with a **Facilitator Note** written into the build — the common misunderstandings, what a finished submission looks like, and an early-finisher extension. Read it before the lab week; it is your answer key for that lab.

Before publishing each VR page: the “How to Join” block ships with **[SESSION ID]** and **[LOCATION NAME]** placeholders. Create the persistent session, then fill both in on all twelve S1 VR pages (6 activities + 6 labs). A page published with the placeholder is the most likely support ticket of Week 1. Students who cannot get into VR are entitled to an alternative path — the standing fallback is desktop-mode participation, or the same artifact built as a labeled paper/whiteboard diagram photographed and submitted with the same reflection, graded on the same rubric.

SECTION 4

4. Using Ignite AI to support students (teacher-facing)

[VERIFY] Ignite AI’s exact feature names and availability vary by Canvas release and by what Optima has enabled. Confirm the current feature set in Optima’s Canvas instance before leaning on a specific button. Everything below is described functionally.

You are facilitating roughly 500 students. The Canvas AI assistant is how the arithmetic stays humane — used under one standing rule: **AI drafts, the teacher decides**. You read, edit, and own everything that reaches a student or family. AI is never used to surveil, rank, or auto-penalize.

- **Draft individualized feedback.** For VR lab grading, have it draft a two-sentence comment from your rubric clicks; edit for accuracy and voice, then send. This is how 500 rubric comments stay personal at 2–3 minutes each.
- **Summarize wrapper analytics.** Paste or point it at the outro’s “what remains unclear” export and ask for the five most common sticking points with representative quotes. Verify against your own skim, then reteach the real winner.
- **Surface students who need a nudge.** Ask for the intersection of low outro confidence, missing assignments, and declining quiz scores — then *you* write (or edit) the warm check-in each flagged student gets. The flag finds them; the human reaches them.
- **Generate practice variants.** A student who bombed the binomial quiz gets three fresh parallel problems with worked solutions — drafted in seconds, checked by you before sending (verify the arithmetic; this is mathematics, and AI-generated numbers are guilty until proven correct).
- **Communicate with families.** Draft clear, kind progress summaries and translations. Same rule: read, edit, own.

SECTION 5

5. Grading categories & Canvas gradebook setup

The built Semester 1 point inventory: 51 assignments × 10 pts = 510; 51 quizzes × 5 pts = 255; 6 VR activities × 5 = 30; 6 VR labs × 15 = 90; 6 outro wrappers × 2 = 12; review activity 10; final exam 50. **Semester 1 total: 957 points.** The built Semester 2 inventory has the same shape: 51 assignments × 10 pts = 510; 51 quizzes × 5 pts = 255; 6 VR activities × 5 = 30; 6 VR labs × 15 = 90; 6 outro wrappers × 2 = 12; review activity 10; final exam 50. **Semester 2 total: 957 points.** Rather than let raw points set the weights, create weighted assignment groups:

Canvas assignment group	Contains	Suggested weight	Tier
Lesson Assignments & Quizzes	All weekly assignments (10 pts) and lesson quizzes (5 pts); the finals-week review activity	50%	1
VR Labs, Activities & Reflections	VR labs (15), VR activities (5), outro wrappers (2)	30%	1–3
Semester Final	Semester Final Exam (50)	20%	1 (auto)

[VERIFY] These weights match the syllabus tiles as shipped; confirm or adjust per department policy *before* the term starts, and keep syllabus and gradebook in agreement. The intro wrappers stay ungraded (0-pt survey) and outside the groups above.

Setup steps (Canvas)

- 1. Course → Assignments → group menu: create the three groups above; enable *Weight final grade based on assignment groups* and enter 50 / 30 / 20.
- 2. File every item into its group as you build it in Canvas (assignment pages and quiz files name their points; the finals spec names the exam).
- 3. Set the outro wrappers as **graded surveys** worth 2 pts (completion credit); intro wrappers as ungraded surveys.
- 4. Attach the native Canvas rubrics from Section 6 to every assignment and VR lab so SpeedGrader scores by click.
- 5. Set the late policy to match the syllabus: no per-item late penalty, 10% deduction in a quarter’s final 14 days, zero after quarter close. (Configure the 14-day deduction manually or by announcement-plus-adjustment — Canvas’s automatic late policy cannot express it directly. [VERIFY] current school practice.)
- 6. Publish module by module, intro wrapper first, outro last, VR pages only after Session IDs are filled in.

SECTION 6

6. Rubrics for every teacher-graded item

Only three rubric shapes exist in this course, deliberately, and they never drift between modules. Every one is already printed on the relevant student page; recreate each once as a **native Canvas Rubric** and attach it everywhere the item type appears.

6.1 Weekly assignment rubric (10 pts — all 51 assignments)

Criterion	Full credit	Partial	Minimal	Pts
Completeness	Every part attempted in full	One part missing or clearly rushed	Two or more parts missing	6
Work shown & reasoning	Steps visible; answers traced to counts/rules; Desmos checks reported	Answers present but audits unreported or vague	Bare numbers, little visible reasoning	4

These are practice sets: grade on complete, honest work, *not* correctness of every number. A neat tree with one wrong number earns full credit; that is stated to students on every assignment page, so hold to it.

6.2 VR lab rubric (15 pts — all 12 labs across the year)

Criterion	Full credit looks like	Pts
Completeness	Every required element of the scene built/annotated; snapshot or recording <i>and</i> reflection both submitted	5
Accuracy	The mathematics in the scene is right (correct probabilities, labels, verdicts — each lab page's rubric row names its specifics)	5
Evidence of engagement	Annotations in the student's own words; scene genuinely arranged and walked; reflection points at specific parts of the scene	5

Each lab's page carries this rubric with lab-specific descriptors plus a Facilitator Note naming what full credit looks like for *that* lab — the note is your per-lab answer guide. Artistic quality is never a criterion.

6.3 VR activity & wrapper scoring (completion)

VR activities (5 pts) and outro wrappers (2 pts) are completion-scored: full credit for a sincere attempt (the artifact is present and on-topic), zero for missing or token submissions. No partial fussing — these exist for formation and signal, not discrimination.

6.4 Free-response quiz questions

There are none, by design: all 255 Semester 1 quiz items are auto-graded (multiple choice, with occasional matching/numeric/true-false), because every item doubles as a final-exam bank item. The only free text you read lives in the wrappers (completion credit) and assignment work. No free-response answer guides are therefore needed — if a future revision adds open-response items, add a 2–3 point analytic guide here first.

SECTION 7

7. Assessment catalog: how to run & grade each item type

Type	What it is & how to grade it	Instances (Semester 1)
Lesson quiz (5 pts, auto)	Five standalone items per lesson; auto-graded. Your job is upstream: file every question into its module bank at creation (Section 8), and watch item analytics for a question missed by >60% of the cohort — that is a reteach flag, not a student flag.	51 quizzes, weeks 1–17 (three per week; see each week's lesson-quizzes.md)
Weekly assignment (10 pts)	Notebook photo/scan or typed entry. Batch in SpeedGrader with the 6/4 rubric — a completeness scan, not a re-derivation; ≈30–45 seconds each once you know the shape. Drop fuller comments only where the work signals confusion.	51 assignments, weeks 1–17
VR activity (5 pts, completion)	Snapshot artifact. Mark complete/incomplete in one look. Common issue: unreadable snapshot angle — return for resubmission rather than penalizing.	M1 Space of Possibilities (Wk 1) · M2 Urn Draws at Scale (Wk 5) · M3 Tree Walk (Wk 7) · M4 Distribution Floor (Wk 10) · M5 Fair-Price Booth (Wk 14) · M6 Two Schools Hall (Wk 16)
VR lab (15 pts, rubric)	The module's deep item. Read the page's Facilitator Note first; grade with the 5/5/5 rubric; write one sentence naming the strongest thing in the scene and (if needed) the next step. The reflection question is the rubric's heart — generic reflections cap Engagement at partial.	M1 Monty Hall at Scale (Wk 3) · M2 Counting Gallery (Wk 6) · M3 Screening Ward (Wk 9) · M4 Galton Drop (Wk 12) · M5 Risk Room (Wk 15) · M6 Betting Parlor (Wk 17)
Intro wrapper (ungraded survey)	No grading. Skim Q1 (essential question) for the misconception census; Q4 gives your cohort baseline; the group-work opt-in feeds any collaborative pairings you offer.	First item of each module
Outro wrapper (2 pts, graded survey)	Completion credit; the value is the analytics triage (Section 2).	Last item of each module
Review activity (10 pts)	Finals-week mixed practice; grade like a weekly assignment.	Week 18
Semester final (50 pts, auto)	Bank-draw exam; see Section 8.	Week 18
Discussions	<i>This course has no separate discussion forums.</i> The module's Tier 2 reflection load is carried by the outro wrapper instead — a deliberate choice that keeps the weekly cycle clean. If your section wants community discussion, run it in office hours / study hall rather than adding a graded forum.	—
Excel labs	Semester 2 only: workbook uploads with complete click-path instructions in the lesson. Grade on the assignment rubric; open the workbook, check the formulas exist and compute. The eleven uploads are Assignments 23.3, 25.3, 26.3, 27.3, 28.2, 29.1, 30.2, 31.1, 32.2, 33.3 and 34.1 (named like Wk27_Prediction_YourName.xlsx). Where each student's random numbers differ (Weeks 30–32), grade structure and consistency, not identical values. [VERIFY] The Excel click-paths	Wks 23, 25–34

were written without being tested in a live copy of Excel; collect student defect reports in the module outro wrappers.

SECTION 8

8. The semester finals

Each semester final is a 50-question, 50-point, 90-minute Classic Quiz built entirely from that semester's lesson-quiz banks — **no new items, ever**. That promise is made to students all semester ("every quiz question also feeds your final exam bank") and is the fairness rationale: preparation reduces to honest review of one's own quiz history. The full implementation spec lives in `semester-1-finals/finals-quiz-spec.md`; the short version:

- Six question banks, named PSH S1 · Module N Bank (45/45/45/45/45/30 items). **File every lesson-quiz question into its bank at creation time** — do not defer to finals week.
- The exam is six question groups, each linked to its bank, drawing 9/9/9/9/9/5 at 1 pt each. Every student gets a different, proportionally comprehensive form.
- Settings: shuffle answers ON, 90 minutes, 1 attempt, one-question-at-a-time ON, backtracking teacher's choice (decide before publishing), correct answers shown after the due date.
- Do not edit bank items while the exam window is open.

Semester 2. The Semester 2 spec is `semester-2-finals/finals-quiz-spec.md`: six banks named PSH S2 · Module N Bank (N = 7–12; 45/45/30/45/45/45 items), six question groups drawing 9/9/6/9/9/8 at 1 pt each. It includes a standalone audit of all 255 items. One finding to act on: the correct answer sits in position a for 51 items, b for 86, c for 76 and d for 42, so keep **shuffle answers ON**.

SECTION 9

9. Weekly announcement scripts

In an asynchronous course the weekly announcement is your classroom presence. Post every Monday morning. Clone this template, then use the module scripts below as drafted starting points — edit into your own voice.

TEMPLATE — [Week N]

Scholars — this week you're in [module / lessons]. The heart of the week is [one concept, one sentence]. Don't miss [the one thing: a VR lab, a tricky quiz, a due assessment] by [due date]. [One-line encouragement or a one-line puzzle]. — [Name]

MODULE 1 OPENER (Week 1)

Welcome to Probability & Statistics Honors! Start with the Module 1 "Begin Here" page, then meet the three doors that fooled professors of mathematics. Write your answer down before you open the reveal — being confidently wrong on page one is the whole point, and the feeling has a name here: wonder. Have [desmos.com/scientific](https://www.desmos.com/scientific) bookmarked; your first VR activity (The Space of Possibilities) is open now.

MODULE 2 OPENER (Week 4)

Module 2 begins: the rules of the game. This module rewards exactness — the multiplication and addition rules, and in Week 6, counting that gets big fast (why a longer password beats a fancier one). Perseverance is the thread: the problems punish shortcuts and reward the scholar who names every case.

MODULE 3 OPENER (Week 7)

Module 3 is the deepest water of the semester: conditional probability and Bayes' theorem — the mathematics of honestly changing your mind. The Screening Ward VR lab in Week 9 will put you inside the most misunderstood calculation in medicine. If trees gave you trouble in Module 1, this is the module to visit office hours early.

MODULE 4 OPENER (Week 10)

Module 4 turns chance into a number you can weigh: random variables, distributions, and expected value. By Week 12 you'll drop a thousand balls down a Galton board and watch disorder assemble a bell curve — the single most beautiful sight of the year. Discipline is the thread: name every outcome, weight every branch.

MODULE 5 OPENER (Week 13)

Module 5 asks the year's most practical question: how should you decide when you can't be sure? Expected value meets insurance, fair prices, diversification — and then two famous paradoxes that test whether the theory describes us. The Risk Room lab has you walk a decision with real stakes drawn in.

MODULE 6 OPENER (Week 16)

Our shortest, strangest module: what does "probability" even mean? Two schools of thought have argued about it for a century, and the argument matters for every statistic you'll read next semester. Intellectual honesty is the thread. The Betting Parlor lab will measure your own beliefs — and check whether they're even coherent.

FINALS WEEK (Week 18)

Finals week: the review lesson ties the semester's six rooms into one machine, the review activity is your warm-up, and then the exam — 50 questions, 90 minutes, every one drawn from quizzes you've already taken. Review your own quiz history; that is the study guide. Desmos open in another tab, pencil and paper at hand. Well earned, scholars.

MODULE 7 OPENER (Week 19)

Statistics begins with data, and the thread is honesty. This module teaches you to display data truthfully and to catch the graph that is lying. Your first Excel workbook arrives in Week 21; keep Excel open beside Desmos from now on. The Data Safari VR activity is open now.

MODULE 8 OPENER (Week 22)

Module 8 is about summarizing a pile of numbers fairly: center, spread, shape, and what an outlier is really asking you. Courtesy is the thread: say clearly what the data say, and what a single summary hides. Walking the Distribution closes the module in VR.

MODULE 9 OPENER (Week 25)

Two short weeks on one beautiful idea: the normal curve, and when it fits and when it does not. Wonder is the thread: watch order rise out of randomness on the Galton board in Week 26, then ask when the bell should not be trusted. The z-Score Lineup VR activity is open now.

MODULE 10 OPENER (Week 27)

Module 10 asks how two variables move together, and what we are allowed to say about it. Fit a line, read its residuals, and keep one sentence at hand all module: correlation is not causation. Responsibility is the thread: careful claims. Bring your Excel skills; several workbooks are due.

MODULE 11 OPENER (Week 30)

Where does data come from? Module 11 is about samples and study design: who you ask, what you ask, and how treatments are assigned. Responsibility again, because whoever chooses the sample holds the power to tilt the answer. The Sampling Orchard lab in Week 32 puts you among the trees.

MODULE 12 OPENER (Week 33)

The last module turns everything into inference: how far a sample can be trusted, margins of error, confidence intervals, comparing groups, and learning to critique a report. Service is the thread: people will act on the numbers you give them, so report uncertainty honestly. Week 35 ends with a full written critique.

FINALS WEEK (Week 36)

Finals week: the review lesson walks the semester from one sample to a population, the review activity is your warm-up, and then the exam: 50 questions, 90 minutes, every one drawn from quizzes you have already taken. Review your own quiz history; that is the study guide. Desmos and Excel open, pencil and paper at hand. Well earned, scholars.

SITUATIONAL — mid-module check-in

Quick check, scholars: if the last look-back left you shaky — or you haven't said so out loud yet — office hours are [day/time] and my inbox is open. In this course confusion is data, and reporting it honestly is the virtue we grade ourselves on.

SITUATIONAL — pre-VR-lab

The [lab name] VR lab closes out this module. Review your notebook [diagram/table] before you put the headset on — it's your script in there. Charge the headset tonight; if you can't get into VR, message me before the deadline and we'll set up the alternative path.

SITUATIONAL — re-engagement nudge (individual message)

Hi [name] — I noticed nothing has come in from you since [date], and I wanted to check in before the quarter math starts working against you. Which lesson are you on? If something's in the way — schedule, tech, the material itself — tell me and we'll make a plan. You have time to be fully back on track this week.

10. Teacher workflow & workload map

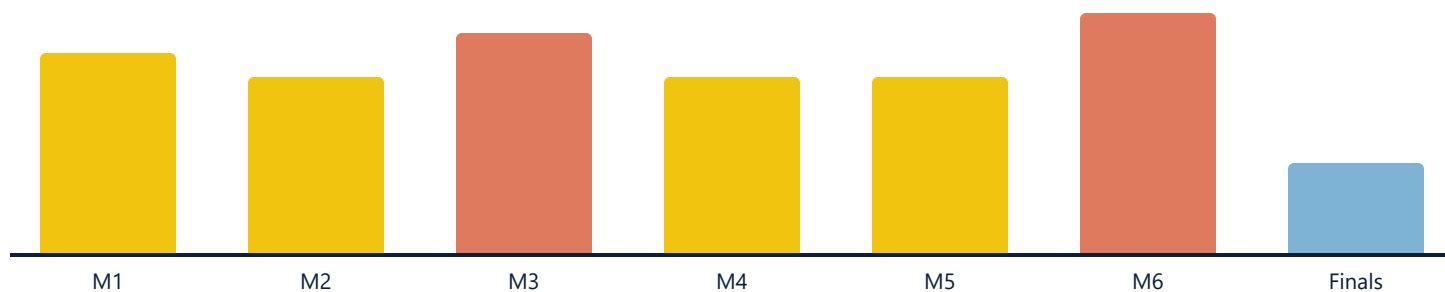
The normal week, in order

1. **Monday:** post the announcement; scan last week's quiz item analytics for cohort misses (10–15 min).
2. **Midweek:** batch-grade the week's assignment submissions in SpeedGrader with the 6/4 rubric; comment only where work signals confusion.
3. **Module-close weeks:** triage the outro wrapper (15–20 min), then message flagged students; grade the VR lab across the following days, front-loaded by the page's Facilitator Note.
4. **Ongoing:** answer messages; hold office hours / study hall / VR workshop.

Where the heavy weeks are

Grading load is flat-ish week to week (assignments + quizzes), with a bump at every module close (VR lab rubric + outro triage) — weeks 3, 6, 9, 12, 15, 17. Week 17 is the year's heaviest: Module 6 closes (lab + outro) *while* you verify the six question banks and build the final. Week 18 itself is light (the exam auto-grades) — spend it on bank verification, exam previews, and re-engagement outreach.

Semester 2 follows the same rhythm, with module closes in weeks 21, 24, 26, 29, 32 and 35. Two stretches run heavier than the Semester 1 pattern because they carry teacher-graded written work in addition to the VR lab: Weeks 32–33 (the Assignment 32.3 design task) and Week 35, the year's heaviest (Module 12 closes with its lab and outro while the Assignment 35.3 capstone critiques come in and you verify the six Semester 2 banks and build the final). Week 36 is light, as Week 18 is.



Relative facilitator load by module, Semester 1. Blue = light, gold = moderate, coral = heavy. M3 runs hot on student-help volume (conditional probability & Bayes); M6 compresses a module close into two weeks against finals prep. Heights reflect the tiered item counts (each module: ~9 assignments, ~9 quizzes, 1 lab, 1 activity, 2 wrappers; M6: 6/6/1/1/2).

SECTION 11

11. Coach, not catch

Your job is to move each scholar forward, not to detect and punish. In practice:

- Feedback names the next step, not just the error — and leads with what is working in the submission.
- Assignments are graded on complete, honest work by design. Honor that: a wrong number with visible reasoning is a teaching moment, not a deduction hunt.

- Use the wrapper analytics and any AI assistance to **find students who need help early**, then reach out warmly — never to build a case.
- On suspected AI misuse or dishonesty (the syllabus defines AI-generated filler as an insincere attempt): open a conversation first. Ask the student to walk you through their tree, their table, their reasoning — in this subject, five minutes of talk-through settles authenticity better than any detector, and treats the moment as formation.
- Public presence (announcements, office hours) models curiosity and names good thinking by name; it never singles out weak work.

SECTION 12

12. What makes this course classical & cultivating culture

The classical character is in the build, not the vocabulary:

- **Wonder before mechanism.** Nearly every lesson opens with a problem that surprises (Monty Hall, the streak that isn't due, the screening test that's usually wrong) and requires the student to *commit an answer in writing* before the reveal — the classical practice of confronting one's own ignorance honestly.
- **Imitation of worked examples.** Every procedure is modeled completely, then imitated on parallel problems — worked-then-independent, the classical *imitatio* pattern.
- **The Great Conversation.** History-of-mathematics callouts run through the year — Pascal and Fermat's letters, Bayes' posthumous essay, Bernoulli, Laplace, Galton's quincunx, Fisher's lady tasting tea — and Module 6 hands students a genuine, unresolved philosophical argument (what probability *is*) as a live question, not a settled fact.
- **Virtue threads that do real work** (confirmed by the curriculum team): M1 Wonder · M2 Perseverance · M3 Honesty · M4 Self-government · M5 Courage/Responsibility · M6 Honesty · M7 Honesty · M8 Courtesy · M9 Wonder · M10 Responsibility · M11 Responsibility · M12 Service. Each module's outro wrapper asks the student to reflect on its virtue specifically — read a few every module; they are the course's soul in the students' own words.

Culture, asynchronously: name good student thinking in your announcements (with permission); answer “still unclear” reports publicly and gratefully so reporting confusion stays safe; keep the virtue thread visible in your own writing; and treat every piece of feedback as relationship — at 500:1, your sentences are the school's face.

SECTION 13

13. Sensitive-content flags & family scripts

Where	What	Framing in the build
Throughout S1 (esp. Wks 3–5, 13–16)	Gambling contexts: casinos, lotteries, betting rates, fair prices of gambles	Gambling is the historical laboratory of probability and the course treats it diagnostically: the Gambler's Fallacy is <i>refuted</i> , expected value shows house games are losing propositions, and the “betting rate” of Module 6 is a measurement device for belief, not an endorsement of wagering.
Week 8–9 (M3)	Medical screening, false positives, disease prevalence examples	Standard base-rate content; handled factually and without graphic material. Students with medical anxiety or family diagnosis experience may find it personal — the mathematics is compassionate here (a positive test usually isn't a diagnosis).
Week 15, Lesson 3	Pascal's Wager — a decision-theoretic argument about belief in God	Presented as decision theory's most famous historical application and examined on its mathematical merits (infinite expected values), with Pascal's <i>Pensées</i> context. Respectful of belief; endorses no theological conclusion.
Week 15, Lesson 2	Daniel Ellsberg history note (decision theorist, later the Pentagon Papers)	Brief biographical callout; historical, not partisan.

FAMILY SCRIPT — send before Module 5 (or at S1 start)

Dear families — a note on content you may hear about at the dinner table. Probability was born at the gambling tables of the 1600s, and our course uses that history the way mathematicians always have: dice, lotteries, and bets appear as examples so that scholars can compute exactly why games of chance are losing propositions — the strongest inoculation mathematics can offer. Later this semester, scholars also weigh medical screening tests (why a positive result usually needs a second look) and meet Pascal's famous “wager,” studied as a piece of decision theory and treated with respect for belief. If you'd like to talk through any of this, I'd welcome the conversation — [name, email, office hours].

SECTION 14

14. Standards coverage map (CPALMS → modules & lessons)

Florida course 1210300 benchmarks, from the CPALMS course export in 01_Analysis/. Codes are tracked internally only — **never shown on student pages**. Lesson references use week.lesson (e.g., 8.3 = Week 8, Lesson 3). Every benchmark is assessed continuously through its weeks' lesson quizzes and paired assignments, and again on the semester final via the module's bank draw; rows note additional deep-assessment sites. Two numbering notes: the course's CPALMS list includes DP.2.9 but not DP.2.8, and DP.6.5/6.7 but not DP.6.6 — those absent codes are not part of this course.

By standard — Data & Probability benchmarks (MA.912.DP)

Benchmark	Short description	Taught (module · lessons)	Deep assessment
DP.1.1	Select an appropriate display for numerical/categorical, univariate/bivariate data	M7 · 19.3–21.1	Wk 21 Excel display lab; M7 VR Chart Room lab
DP.1.2	Interpret data distributions in various displays (spreadsheets/technology)	M7 · 20.3, 21.3	Wk 21 Excel workbook upload
DP.1.3	Correlation vs. causation, numerical and categorical contexts	M10 · 28.3	Wk 28 assignment (confounder analysis)
DP.1.4	Estimate population values from a sample; develop margin of error by simulation	M12 · 33.2	Wk 33 assignment + S2 final
DP.1.5	Interpret margin of error and its confidence level	M12 · 33.3	M12 VR Interval Walk activity
DP.2.1	Compare measures of center/variability across data sets; outlier effects; shape	M8 · 22.1–24.3	Wk 23 Excel stats lab; M8 VR Walking the Distribution lab
DP.2.2	Fit data to a normal distribution; recognize when inappropriate	M9 · 25.1, 26.1–26.3	Wk 26 normal-applications lab; M9 VR Inside the Bell lab
DP.2.3	Estimate population percentages from a normal fit (empirical rule, technology)	M9 · 25.2–25.3, 26.3	Wk 26 lab + S2 final
DP.2.4	Fit a linear function; interpret slope/intercept; predict in context	M10 · 27.1–27.3	Wk 27–28 Excel trendline work; M10 VR Scatter Field lab
DP.2.5	Assess a linear fit with residuals	M10 · 28.1	M10 VR Scatter Field lab (residual pillars)
DP.2.6	Determine correlation strength/direction from scatter, fit, residuals	M10 · 28.2	Wk 28 assignment
DP.2.7	Compute the correlation coefficient with technology; interpret it	M10 · 28.2 (Excel CORREL)	Wk 28 Excel workbook
DP.2.9	Fit an exponential function (log transform, technology)	M10 · 29.1	Wk 29 assignment (log-transform lab)
DP.3.1	Construct a two-way frequency table; joint/marginal frequencies; association	M10 · 29.2	Wk 29 assignment + S2 final
DP.3.2	Construct a two-way relative frequency table from given frequencies	M10 · 29.3 (incl. building a table from marginal and conditional frequencies, with a tree diagram)	Wk 29 assignment
DP.3.3	Interpret joint/marginal/conditional relative frequencies (incl. false positives/negatives)	M10 · 29.3; groundwork in M3 · 8.3, 9.3 (built)	Wk 29 assignment; M3 Screening Ward VR lab
DP.3.4	Construct and interpret a segmented bar graph	M10 · 29.3	Wk 29 assignment

DP.3.5	Solve real-world problems with univariate/bivariate categorical data	M3 · 9.3 (built) ; reinforced M10 · 29.2–29.3	Wk 9 assignment 3 + S1 final (M3 bank)
DP.4.1	Describe events as subsets of a sample space; unions, intersections, complements	M1 · 1.1–1.2 (built)	Wk 1 assignments; M1 VR Space of Possibilities; S1 final
DP.4.2	Determine independence via the product of probabilities	M1 · 3.1; M2 · 4.1–4.2 (built)	Wk 3–4 quizzes/assignments; S1 final
DP.4.3	Calculate and interpret conditional probability	M3 · 7.1–7.3 (built)	Wk 7 assignments; M3 Tree Walk activity; S1 final
DP.4.4	Interpret independence using conditional probability	M3 · 7.3 (built)	Wk 7 quiz 3; S1 final
DP.4.5	Two-way tables as empirical (conditional) probabilities; test independence	M3 · 9.3 (built)	Wk 9 assignment 3; S1 final
DP.4.6	Conditional probability & independence in everyday language	M1 · 3.1–3.3; M3 throughout (built)	Gambler's Fallacy / hot-hand verdicts (Wk 3); S1 final
DP.4.7	Apply the addition rule (mutually exclusive or not)	M2 · 4.2, 5.1 (built)	Wk 4–5 assignments; S1 final
DP.4.8	Apply the general multiplication rule (independent or not)	M2 · 5.2–5.3; M3 · 9.2 (built)	Wk 5 & 9 assignments; S1 final
DP.4.9	Addition & multiplication counting principles	M2 · 6.1 (built)	Wk 6 assignments; M2 Counting Gallery lab; S1 final
DP.4.10	Permutations and combinations	M2 · 6.2–6.3 (built)	Wk 6 assignments; Counting Gallery lab; S1 final
DP.5.1	Parameter vs. statistic	M11 · 30.1	Wk 30 quiz/assignment; S2 final
DP.5.2	Why random sampling is representative	M11 · 30.2	Wk 30 assignment; M11 Sampling Orchard lab
DP.5.3	Compare sampling methods (SRS, stratified, cluster, systematic, judgment, quota, convenience)	M11 · 30.3	Sampling Orchard lab (method comparison)
DP.5.4	Generate multiple/simulated samples to measure variation	M11 · 31.1, 31.3	Wk 31 simulation assignment
DP.5.5	Judge whether a model is consistent with a data-generating process	M11 · 31.3	Wk 31 assignment
DP.5.6	Choose the right design (survey / experiment / observational)	M11 · 32.1, 32.3	Wk 32 design-a-study workshop

DP.5.7	Compare surveys, experiments, observational studies; role of randomization	M11 · 32.1	Wk 32 workshop; S2 final
DP.5.8	Draw inferences about two populations from two random samples	M12 · 34.1	Wk 34 assignment; S2 final
DP.5.9	Compare two randomly assigned treatments; randomization validates causal claims	M11 · 32.2; M12 · 34.2	Wk 34 assignment
DP.5.10	Judge significance of differences using simulation	M12 · 34.3; conceptual groundwork M6 · 17.2 (built)	Wk 34 simulation assignment
DP.5.11	Evaluate data-based reports; misleading displays	M7 · 21.2; M12 · 35.2–35.3	Wk 35 capstone critique
DP.6.1	Define a random variable; graph its probability distribution	M4 · 10.1 (built)	Wk 10 assignments; Distribution Floor activity; S1 final
DP.6.2	Discrete distribution from theoretical probabilities; expected value as mean	M4 · 10.2; M5 · 13.1–13.2 (built)	Wk 10 & 13 assignments; S1 final
DP.6.3	Discrete distribution from empirical probabilities; expected value	M4 · 10.3 (built)	Wk 10 assignment 3; S1 final
DP.6.4	Binomial distributions: expected value, real-world problems	M4 · 11.2–11.3 (built); normal connection M9 · 26.2	Wk 11 assignments; Galton Drop lab; S1 final
DP.6.5	Geometric distributions, real-world problems	M4 · 12.1 (built)	Wk 12 assignment 1; S1 final
DP.6.7	Expected values & standard deviations of decisions; compare strategies; diversification	M4 · 12.2; M5 · 13.1–13.3 (built)	Wk 13 assignments; Risk Room VR lab; S1 final
DP.6.8	Apply probabilities to make fair decisions (lots, random number generators)	M5 · 14.3 (built)	Wk 14 assignment 3; S1 final

By module — quick reference

Module	Primary benchmarks
1 — Foundations of Probability & Logic (built)	DP.4.1, DP.4.2 (intro), DP.4.6
2 — Calculating Probabilities & Counting (built)	DP.4.2, DP.4.7–4.10
3 — Conditional Probability & Bayes (built)	DP.4.3–4.6, DP.3.5; groundwork for DP.3.3
4 — Random Variables & Distributions (built)	DP.6.1–6.5, DP.6.7 (intro)
5 — Decision Theory & Rational Choice (built)	DP.6.7, DP.6.8, DP.6.2 (applied)
6 — Foundations of Inference (built)	Conceptual bridge to DP.5.10–5.11 (no new benchmark introductions; rehearses significance logic)
7 — Data & Data Displays	DP.1.1–1.3 (1.3 intro), DP.5.11 (misleading displays)
8 — Descriptive Statistics	DP.2.1
9 — The Normal Distribution	DP.2.2–2.3 (+ DP.6.4 connection)
10 — Bivariate Data & Regression	DP.2.4–2.7, DP.2.9, DP.3.1–3.4, DP.1.3
11 — Sampling & Study Design	DP.5.1–5.7, DP.5.9
12 — Inference & Statistical Reasoning	DP.1.4–1.5, DP.5.8–5.11

Woven throughout (not mapped to single lessons): the Mathematical Thinking & Reasoning standards MA.K12.MTR.1.1–7.1 (effortful learning, multiple representations, fluency, mathematical discourse, patterns/structure, reasonableness, real-world application) are pedagogy embedded in every lesson's worked examples, widgets, narration activities, and applied problems; ELA.K12.EE.1.1–6.1 (evidence, comprehension, inference, collaboration, format, voice) ride on the course's written verdicts and reflections; ELD.K12.ELL.MA.1 is supported by the multiple-representation design (every concept appears verbally, symbolically, visually, and interactively).

SECTION 15

15. Teaching this course live: the 4-day week

The course was written asynchronous-first, and live sections use the same Canvas content unchanged — the lesson pages become your lesson plans. With four meetings per week and three lessons per week, the natural rhythm is **one lesson per day plus a workshop day**:

Day	Live pattern
Day 1	Lesson 1. Run the page's hook as a whole-class moment (every lesson opens with a commit-before-reveal — collect predictions out loud or on mini-whiteboards <i>before</i> the reveal). Work the worked example on the board while students work alongside; project the widget and let students call the runs. Students finish the activities and start Assignment 1 as classwork.
Day 2	Lesson 2, same shape. Open with a 3-minute retrieval warm-up from yesterday's quiz bank topics. The Calculator Skills callout becomes a 5-minute guided Desmos/Excel drill done together.
Day 3	Lesson 3, same shape. Run the week's AI Training callout as a live demonstration once or twice a module: the class writes the prompt together, predicts, then grades the AI's answer on the projector — the best ten minutes of skepticism training you can buy.
Day 4	Workshop day. Students take the three lesson quizzes (in-class, devices out, Desmos allowed — matching exam conditions), finish and submit assignments, and use remaining time for VR sessions (labs/activities in headset while you circulate), reteach from yesterday's confusion, or extension problems. On module-close weeks this is lab-and-outro day.

Quarter pacing already fits: 8–9 lesson weeks per quarter at 4 meetings/week matches the sample calendar. Keep Canvas as the single source of truth — live students submit the same items to the same gradebook.

Week-by-week live moves (Semester 1)

One suggested whole-class adaptation per week — the moment that converts best from page to room:

Wk	Live move of the week
1	Play Monty Hall physically before anyone reads Lesson 1.1: three folders (one hides a prize card), a student contestant, you as host. Run 15–20 rounds split stay/switch, tally on the board, and let the class's own data start the argument. Then read the lesson together.
2	Build truth tables as a relay: each row of the class fills one row of the table on the board; errors get caught by the next row's student. Ends with the equivalence verdict the lesson proves.
3	The fake-randomness game: half the class writes down an invented "random" 20-flip sequence, half flips real coins and records. Post them anonymously; the class (and you) hunt the fakes by streak length. Sets up the Gambler's Fallacy lesson perfectly — real randomness streaks more than intuition allows.
4	Dice stations: pairs verify the week's multiplication- and addition-rule computations empirically with physical dice, comparing relative frequencies to predictions. Pooling all pairs' data shows convergence.
5	Total-probability urns: two opaque bags with counted tokens; students draw, tally by branch, and reconcile the class totals against the Law of Total Probability computed on the board.
6	Permutations with bodies: four volunteers arrange themselves every possible way while the class counts (24), then "collapse" to combinations by ignoring order (6 groups of 4... choose 2 etc.). The factorial explosion lands physically.
7	Masking-tape probability tree on the floor: students walk paths, multiplying branch probabilities aloud; leaves collect sticky notes that must sum to 1.
8	The 1,000-person grid as a card sort: 100 index cards (or a 10×10 grid on the board) colored by condition and test result for the lesson's screening numbers; the class stands "in the positive corridor" and counts who's actually sick. Live Bayes without the formula, then the formula.
9	Build a two-way table from the room: two quick yes/no questions about the class itself (e.g., plays an instrument / has a sibling), tally live, then compute joint, marginal, and conditional relative frequencies from your own cohort.
10	Carnival day: groups invent a simple dice/spinner game, build its probability distribution, and compute its expected value; the class votes on which game they'd play, then checks the vote against the EVs.
11	Class binomial: everyone flips 10 coins; histogram the class's head-counts on the board and watch the binomial shape appear from ~25 trials. Compare to the widget's thousands.
12	Geometric standing game: all stand, everyone flips until heads and sits when they get it; count who's still standing after each round. The long right tail becomes visible in bodies still on their feet.
13	Insurance role-play: half the room prices a policy (they're the company), half decides whether to buy; both sides compute EV and SD, then negotiate. Diversification demo: one big bet vs. ten small ones, simulated in the widget on the projector.
14	Auction a gamble: sell a real (candy-stakes) gamble to the highest bidder after computing its fair price; the gap between bids and EV opens the utility lesson honestly.
15	Run Allais and Ellsberg as anonymous class votes before the lessons name them; reveal the class's own inconsistency on the board. Nothing sells the paradoxes like discovering you're in one.
16	Two-schools placard sort on the classroom walls (frequency wall vs. belief wall), then elicit betting rates on three fun propositions and check the class's coherence.
17	Reenact Fisher's lady tasting tea with a proxy taste test (e.g., name-brand vs. store-brand). Design the experiment as a class first: how many cups make a claim convincing? That question <i>is</i> significance testing.

18	Finals week: Day 1 review lesson as a guided "grand tour" discussion; Day 2 review activity as team stations; Day 3 open Q&A + Desmos fluency drills; Day 4 proctored final exam (the 90-minute window fits a block; otherwise schedule it Day 3–4).
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Week-by-week live moves (Semester 2)

Wk	Live move of the week
19	Data census of the class (heights, pets, commute minutes, favorite subject): collect once, classify every variable together, and reuse this dataset all semester as "our data."
20	Display sprint: teams get the same small dataset and five minutes per display type (dot plot, stem-and-leaf, histogram, box plot) on big paper; gallery-walk the results and argue which display tells the truth best.
21	Misleading-graph forensics: project real media graphics; teams get two minutes to find the lie (axis, bins, cherry-pick). Then the Excel lab runs as a follow-along build on the projector.
22	Human number line: students holding value cards arrange themselves; the class physically locates median and quartiles, then paces the IQR fence and evicts the outliers.
23	Compute one standard deviation fully by hand as a class (5 values), each student owning one deviation term — then let Excel's STDEV.S confirm it. "Why $n-1$ " becomes a live debate.
24	Dataset showdown: two real datasets (e.g., two teams' season scores), half the room champions each using center/spread/shape evidence; the class judges which claim survived honestly.
25	z-score lineup live: post μ and σ for the class-height data from Week 19; each student computes their own z and lines up in z order. Who is "one standard deviation tall"?
26	Galton board day: run the physical/projected quincunx repeatedly; assign each student a ball to track (unpredictable) vs. the crowd (predictable) — the module's virtue, wonder, does the rest.
27	Guess-the-correlation tournament with the widget on the projector; then teams place a spaghetti line of best fit on a printed scatter before Excel reveals the least-squares answer.
28	Residual detective: give teams a fitted line and data; they build the residual plot and rule on the fit. Then a correlation-headline hunt: each team brings a real "X linked to Y" headline and names three confounders.
29	Log-transform reveal: plot exponential-ish data on normal axes (curved), then re-plot with a log scale live and watch it straighten. Two-way-table speed rounds close the week.
30	Sample the school (or a big jar): estimate a real unknown (beans in the jar works) by different sampling methods per team; post estimates before revealing truth in Week 31.
31	Sampling-variation dot plot: every student's Week-30 estimate goes on one board dot plot — the sampling distribution appears from the class's own work. Then bias-hunt: rewrite three loaded survey questions.
32	Design-a-study workshop as a live pitch day: teams present designs (survey? experiment? observational?), the class attacks with confounders, teams defend or amend. Randomize an actual mini-experiment (e.g., two practice-problem orderings) with RANDBETWEEN.
33	CLT live: each student computes the mean of a random handful from a skewed population (e.g., digit draws); pool the means on a dot plot and watch the bell arrive. Pace the $1/\sqrt{n}$ law down the hallway like the Interval Walk.
34	Simulation court: does the class's experiment from Week 32 show a real difference? Run the shuffling simulation on the projector, place the observed difference, and let the class deliver the verdict.
35	Capstone critique circles: each student brings a data-based news claim; small groups run the full audit (sampling → display → causal language) and present the best takedown — and the best <i>defensible</i> claim, for balance.
36	Finals week: grand-tour review Day 1, stations Day 2, Q&A Day 3, proctored final Day 4.

